CN I (olfactory nerve)

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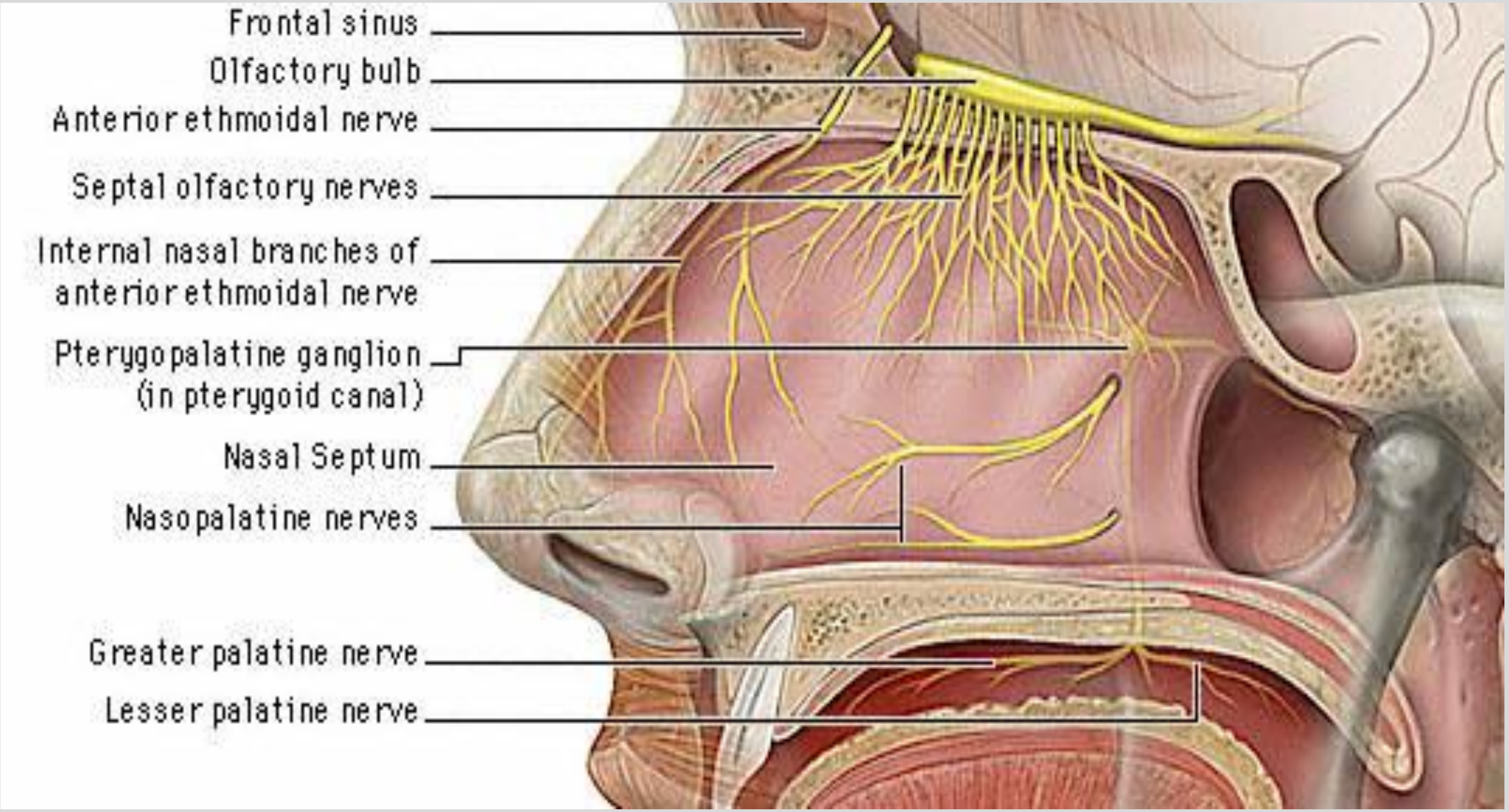
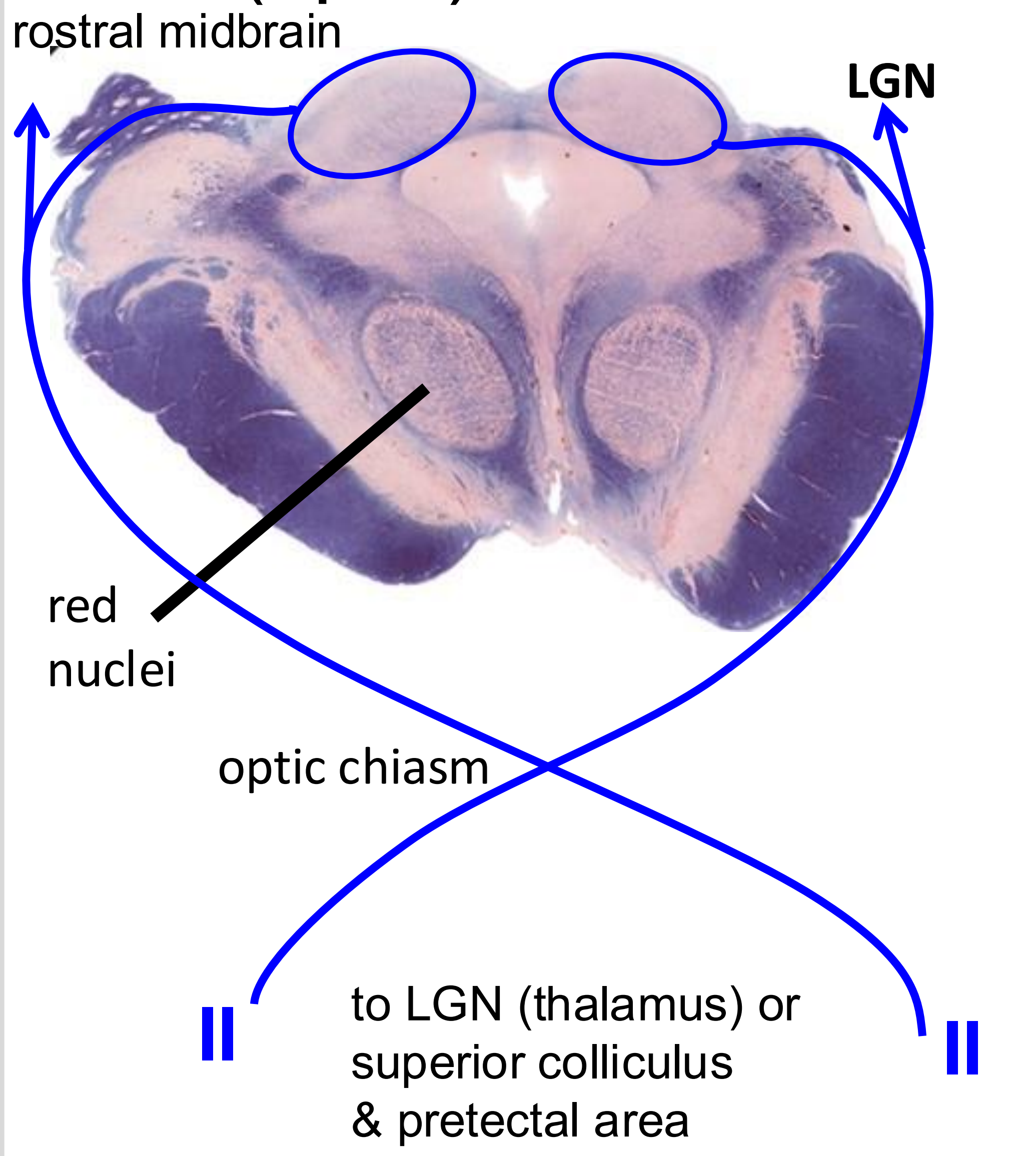
Smell is VISCERAL SENSORY

olfactory epithelium cells → *olfactory bulb* → olfactory tract

the only sensory fibers to bypass thalamus to cerebral cortex



CN II (optic)



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Vision is SPECIAL SENSORY
retina → optic nerves - optic chiasm (<u>some decussate</u>) – optic tract
→ lateral geniculate nucleus of thalamus → occipital lobe

Or

→ superior colliculus

Pupillary light reflex is VISCERAL FEEDBACK
retina → optic nerves - optic chiasm - optic tract → pretectal area → (<u>bilaterally to</u>) Edinger Westphal nuclei

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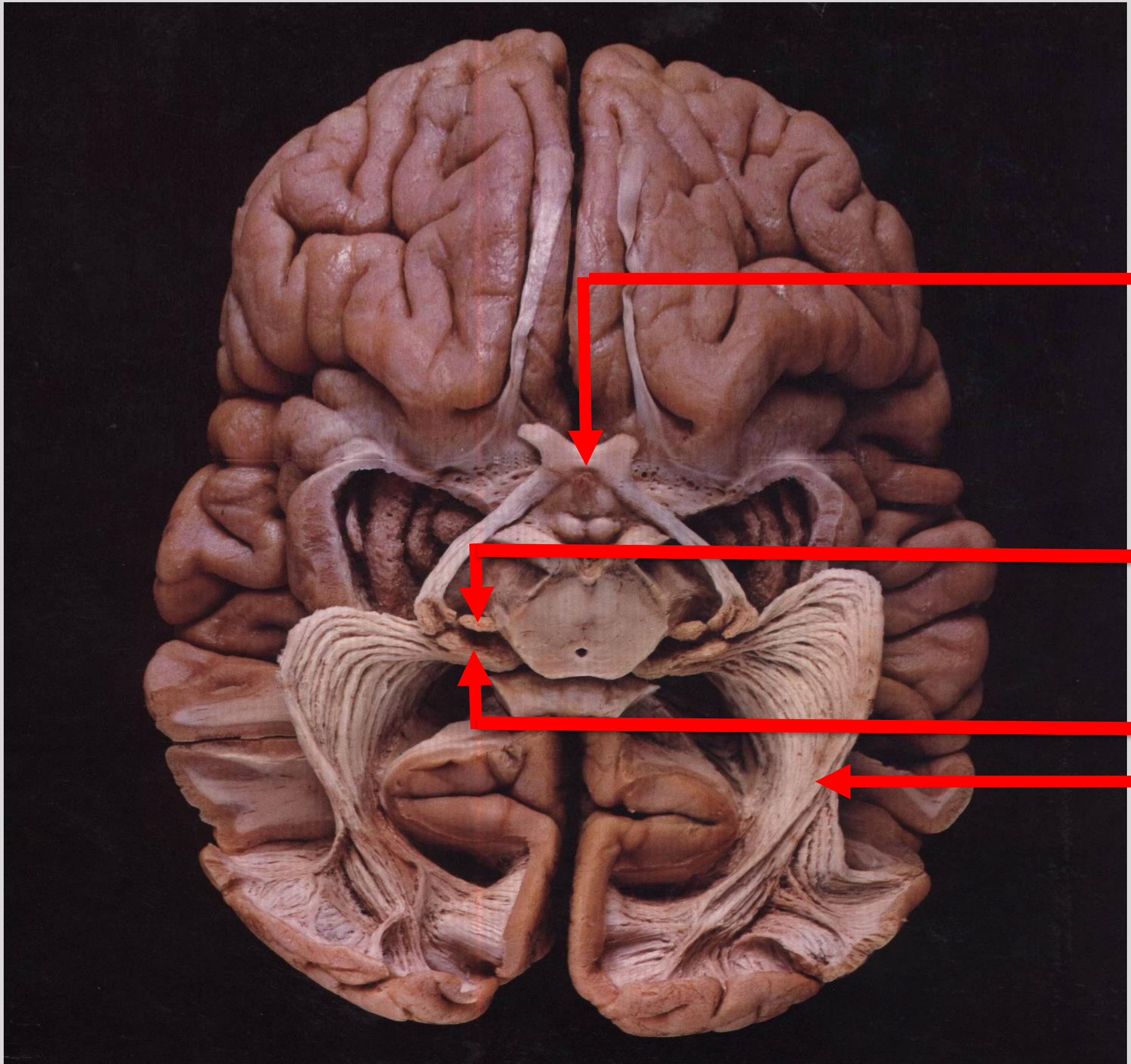
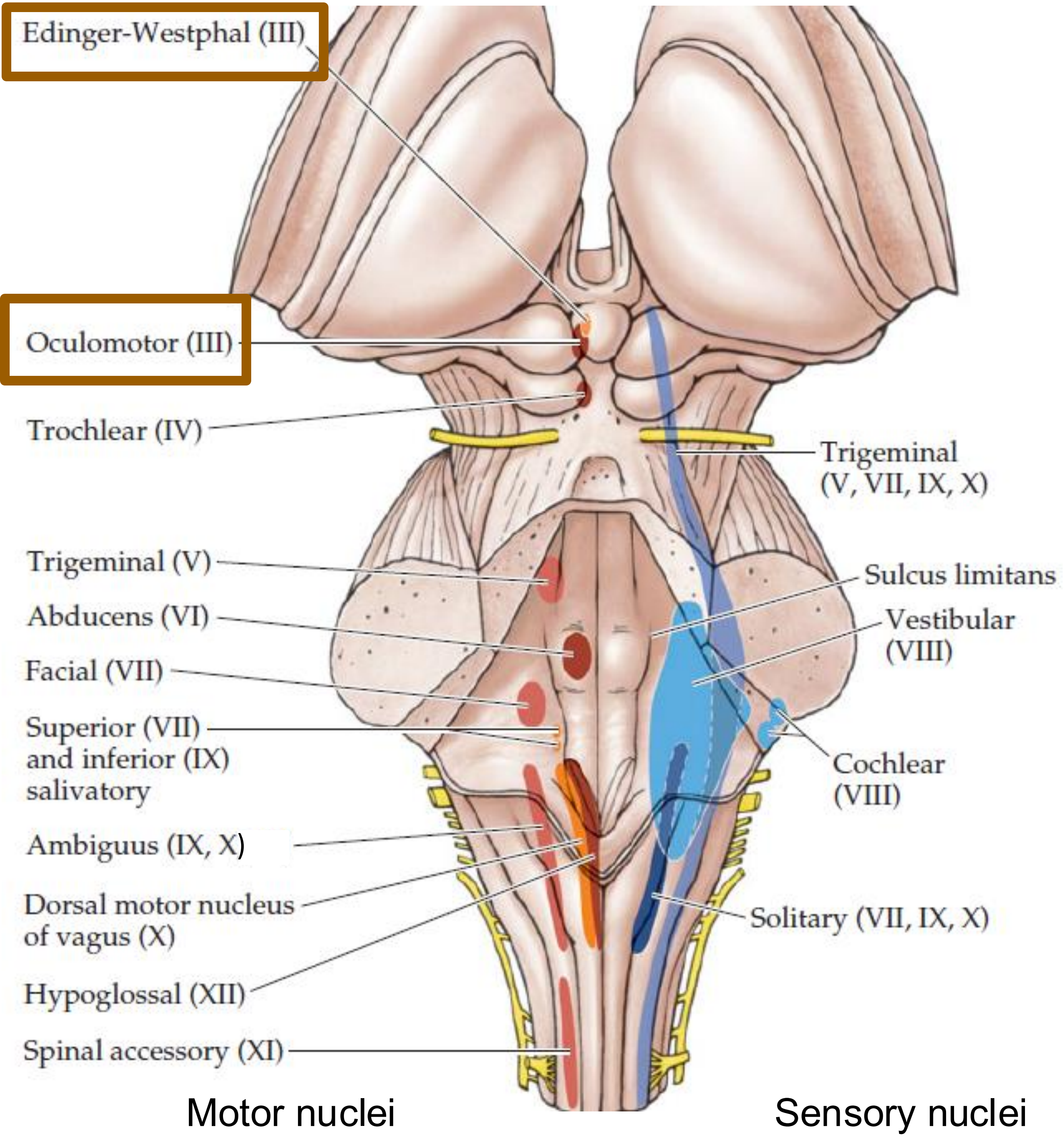
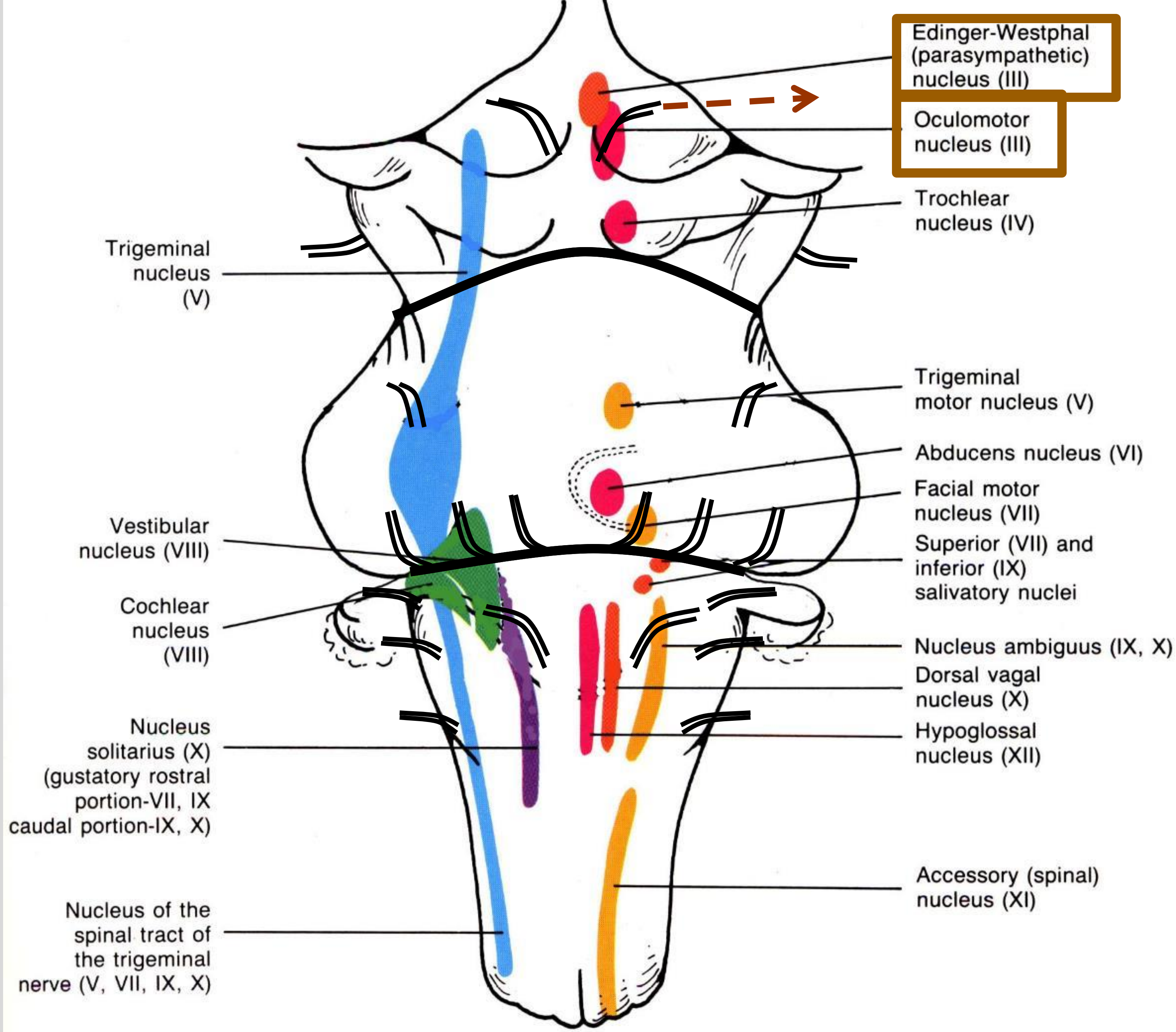


CN III (oculomotor)





Sensory Nuclei

Motor Nuclei

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CN III (oculomotor)

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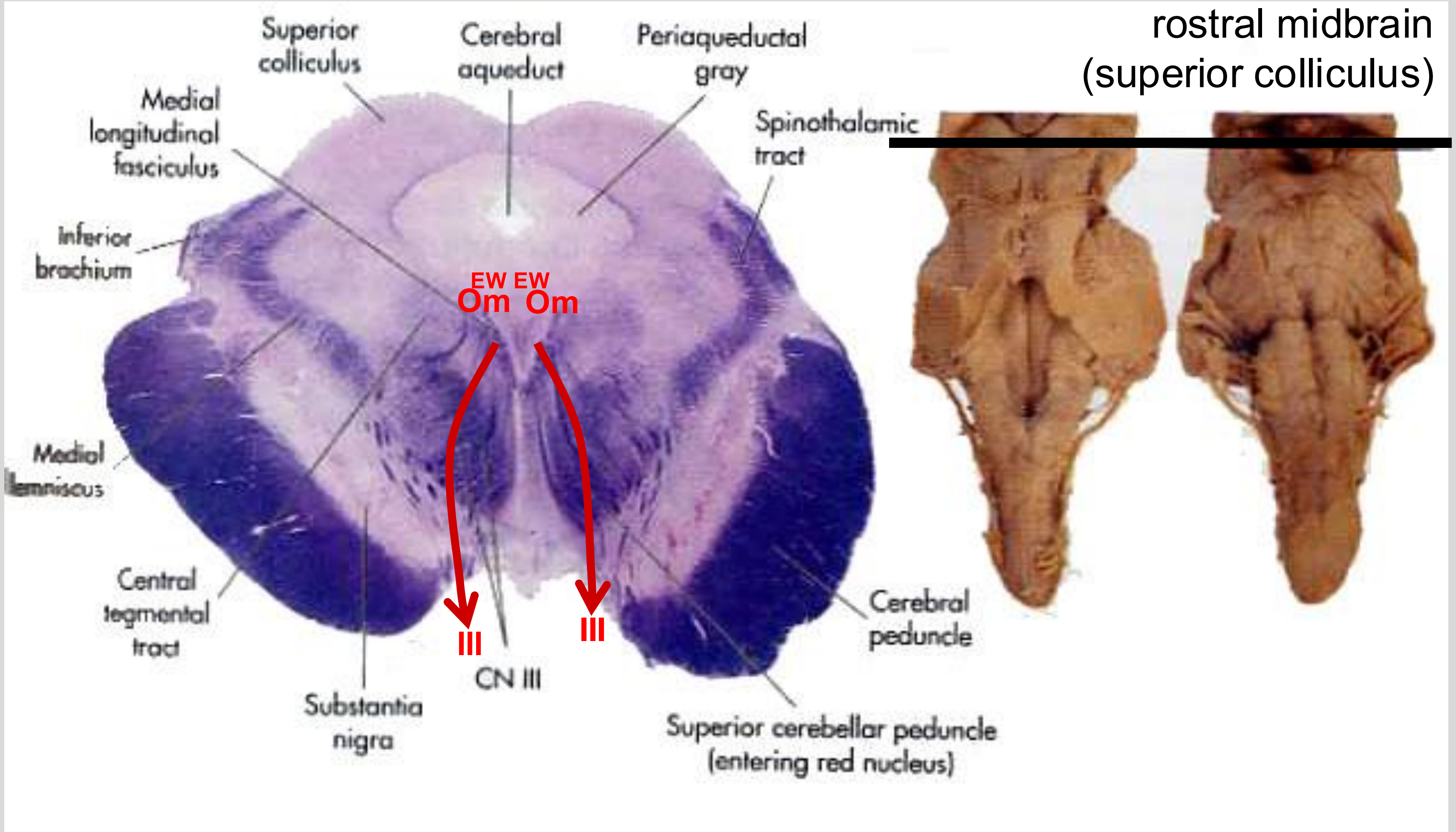
CN III (oculomotor)



III	**Edinger Westphal**	lens & pupil (smooth muscle)
III	**oculomotor**	eye muscles (striated)

SOMATIC MOTOR : ocular positioning and eyelid

oculomotor nucleus →

eyes: medial rectus (ipsilateral) - adduction

inferior rectus (ipsilateral) – mainly depression

inferior oblique (ipsilateral) - extortion (outward rotation)

superior rectus (**contralateral**) – mainly elevation

eyelid: levator palpebrae superioris (**contralateral**)

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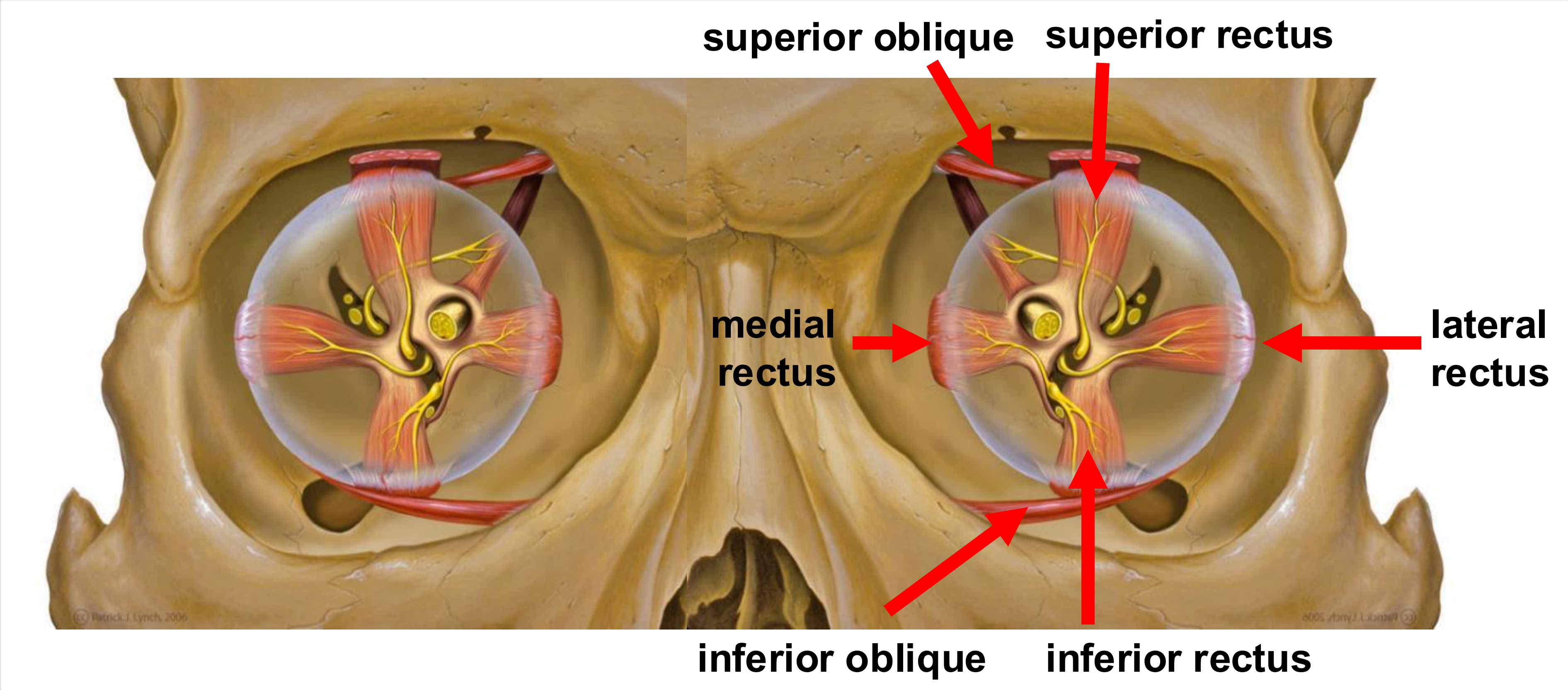
CN III (oculomotor)



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CN III (oculomotor)

SOMATIC MOTOR : ocular positioning and eyelid

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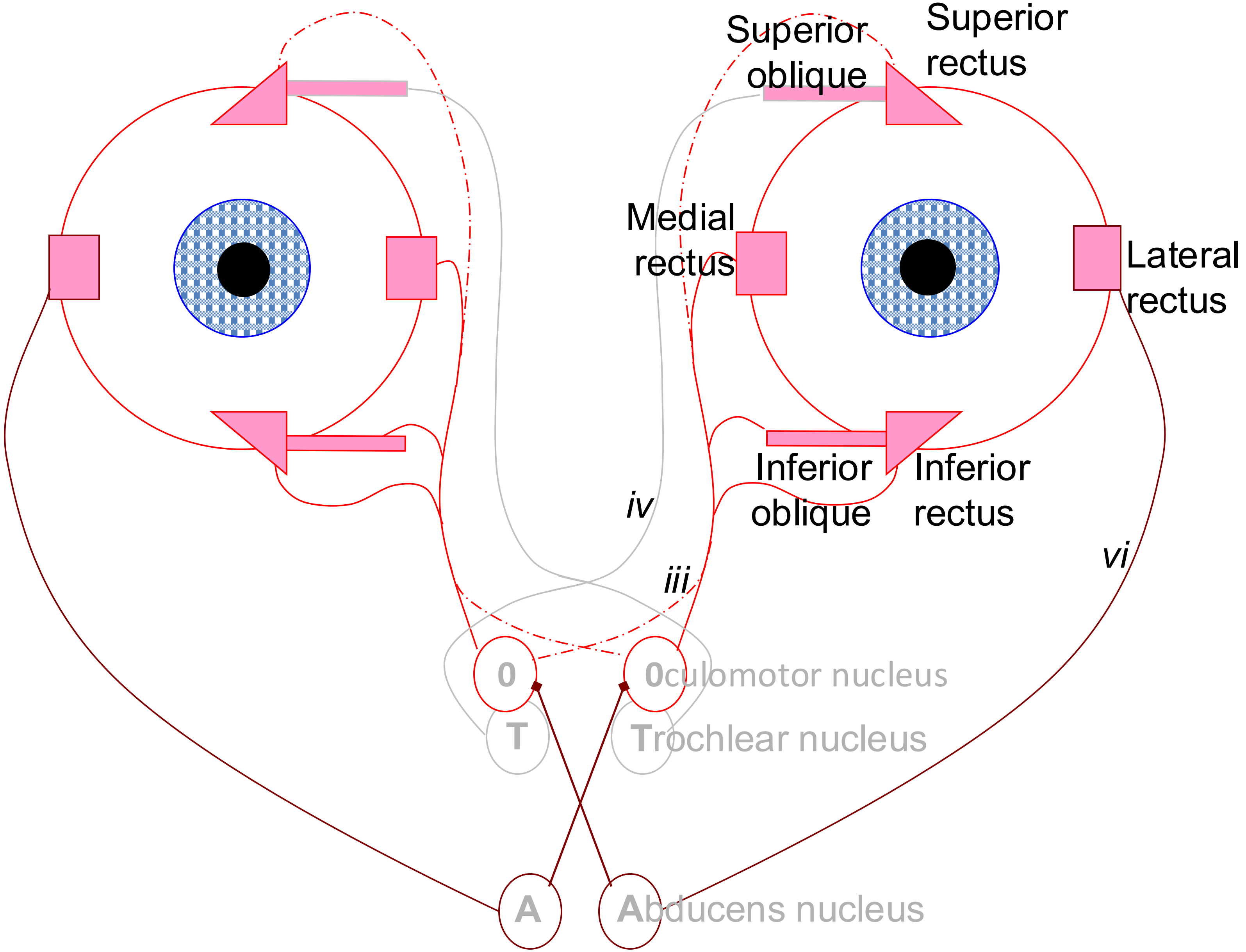
eyelid: levator palpebrae superioris (**contralateral**)

VISCERAL EFFERENT : light reflex

Edinger Westphal nuclei **(part of *<u>oculomotor complex</u>*) →**

pupil contraction →less light: sphincter pupilae (ipsilateral)
lens accommodation* → more convex / shorten focal length: ciliary muscle (ipsilateral)

CN III includes oculomotor nuclei efferents of both sides



Remember - the nerves always control the ipsilateral side

CN III (oculomotor): superior and inferior divisions

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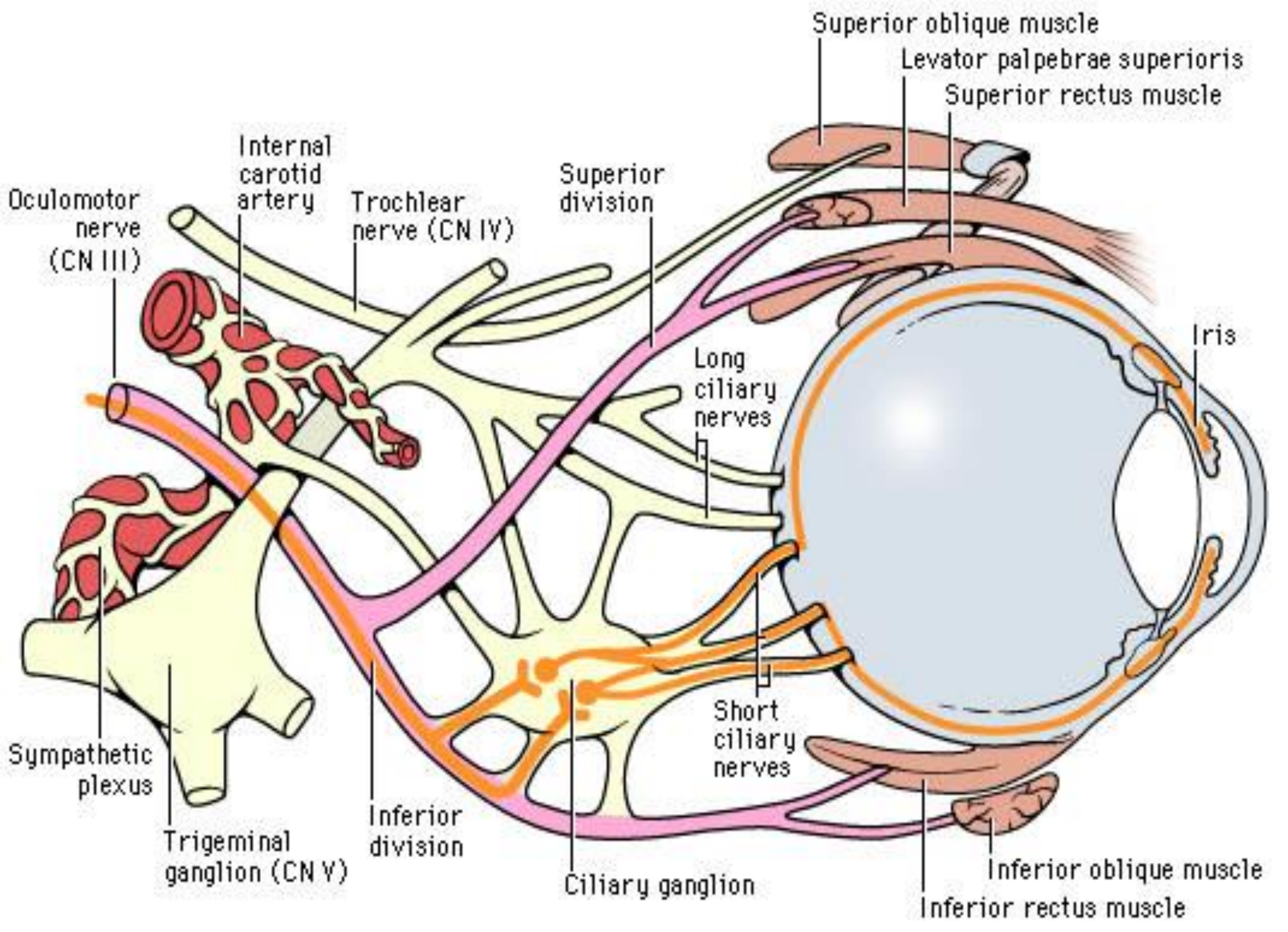

contralateral oculomotor nucleus

superior division
superior rectus &
levator palpebrae superioris

ipsilateral oculomotor and EW nucleus

inferior division
medial rectus,
inferior rectus,
inferior oblique, and
parasympathetics

CN III (oculomotor): brainstem or CN lesion?

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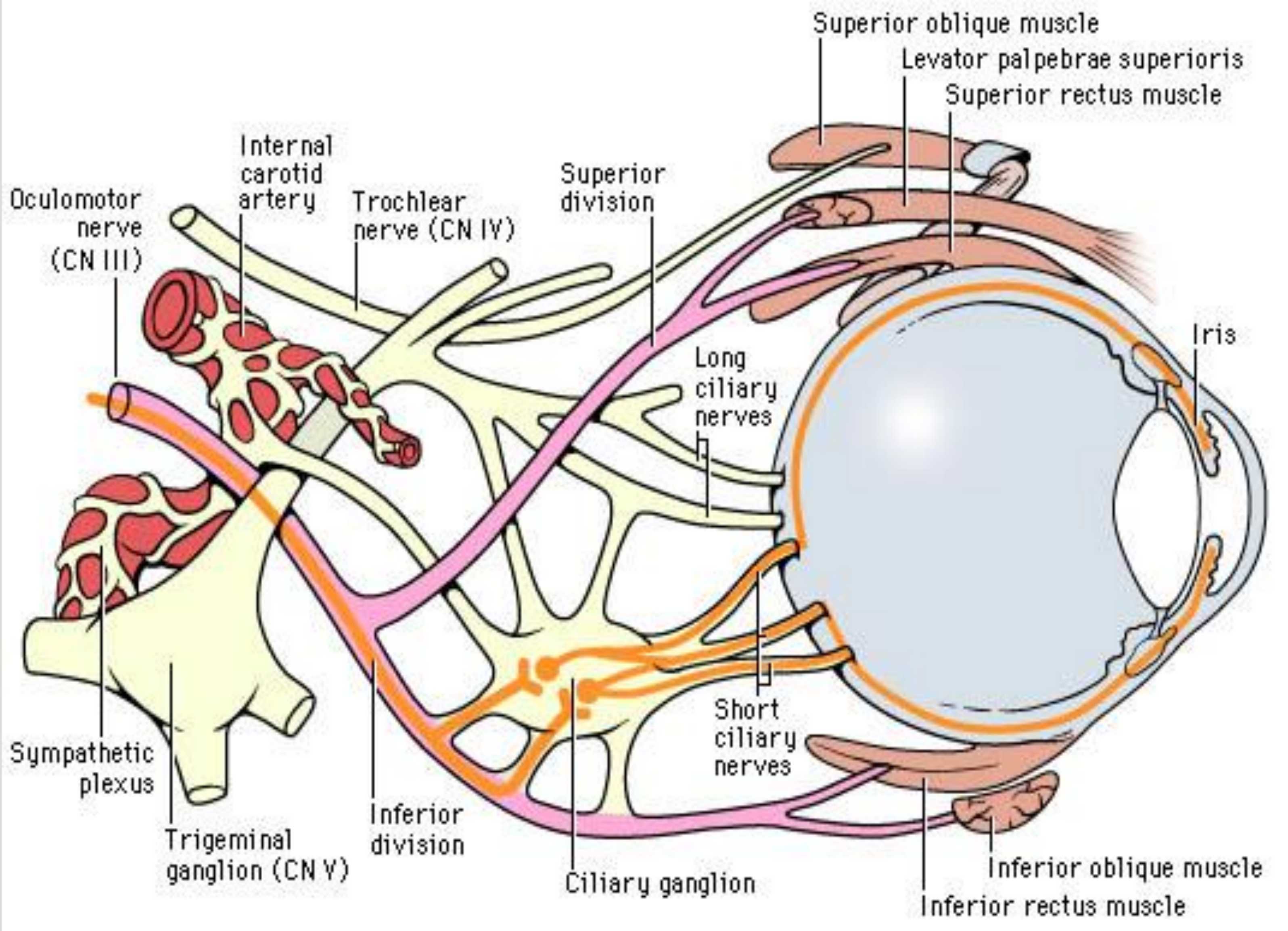

Unilateral brainstem lesions have ipsilateral lower branch symptoms and contralateral upper branch symptoms

Unilateral CN lesions <u>only have ipsilateral symptoms</u>

CN III palsy

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Ptosis

weak or paralyzed levator palpebrae superioris

lesion located in the superior division

Medial rectus palsy

lesion located in inferior division

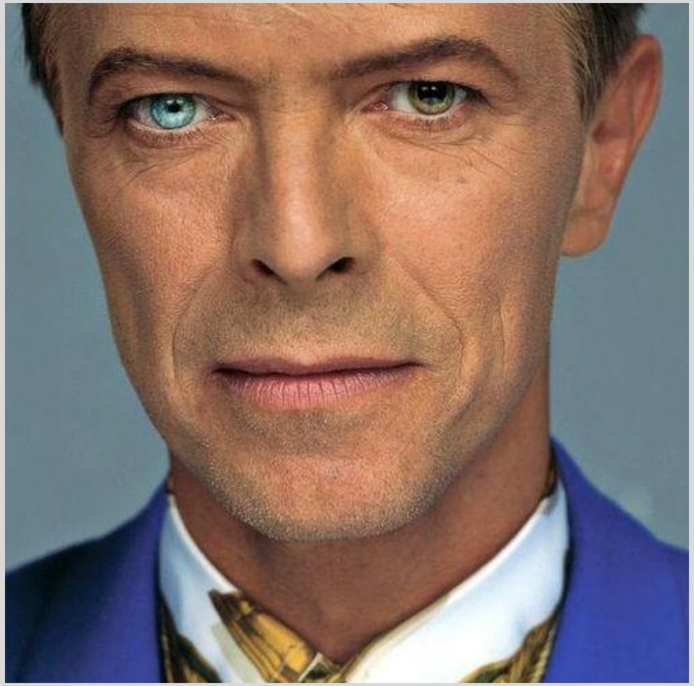
Complete lesion

weakness or paralysis of 4 of the 6 extraocular muscles.

Unopposed muscles pull eye <u>down and out</u>

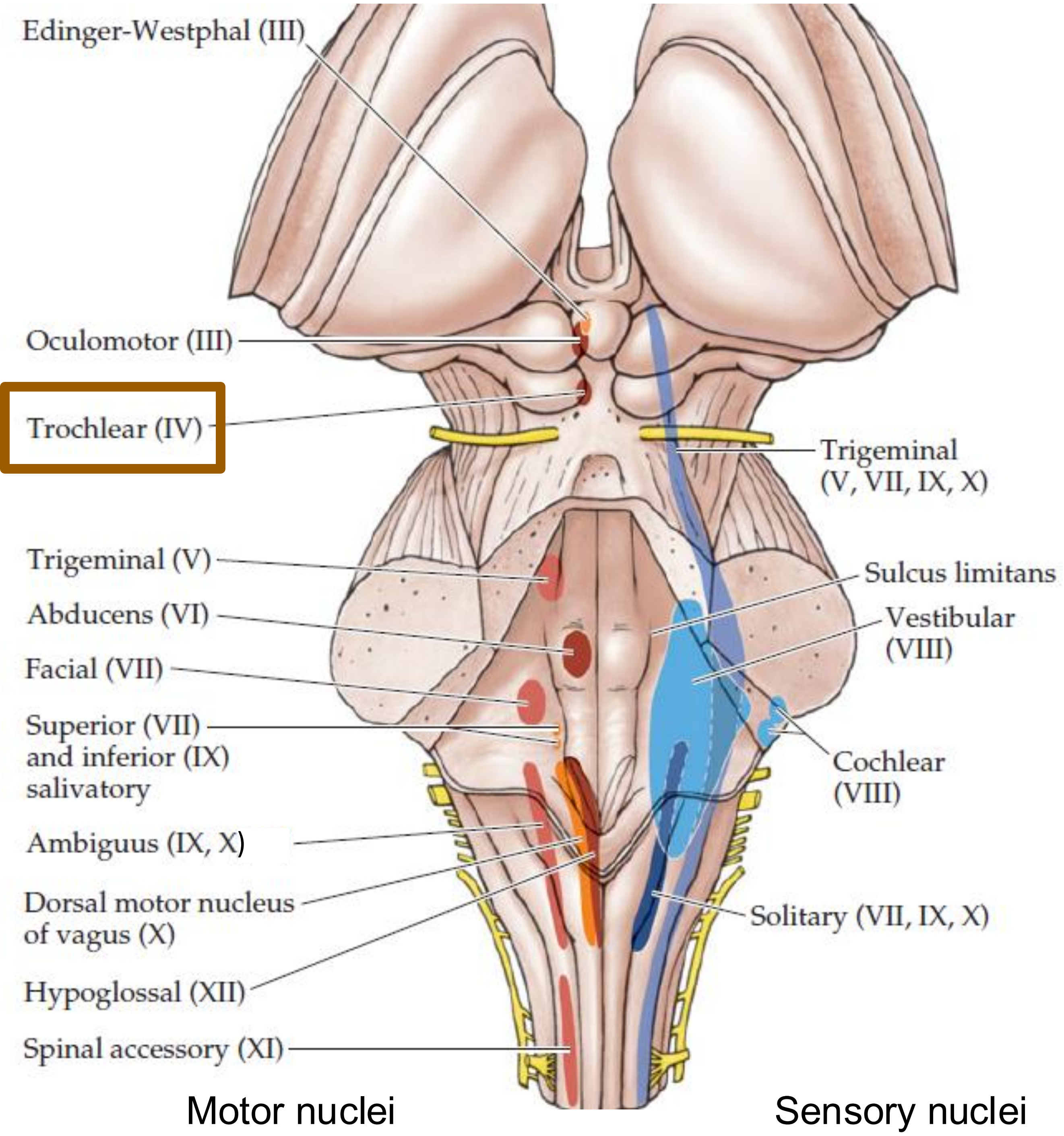
Anisocoria

David Bowie!



Weakness or paralysis of sphincter pupillae. Affected eye remains more dilated than contralateral eye.

lesion located in inferior division



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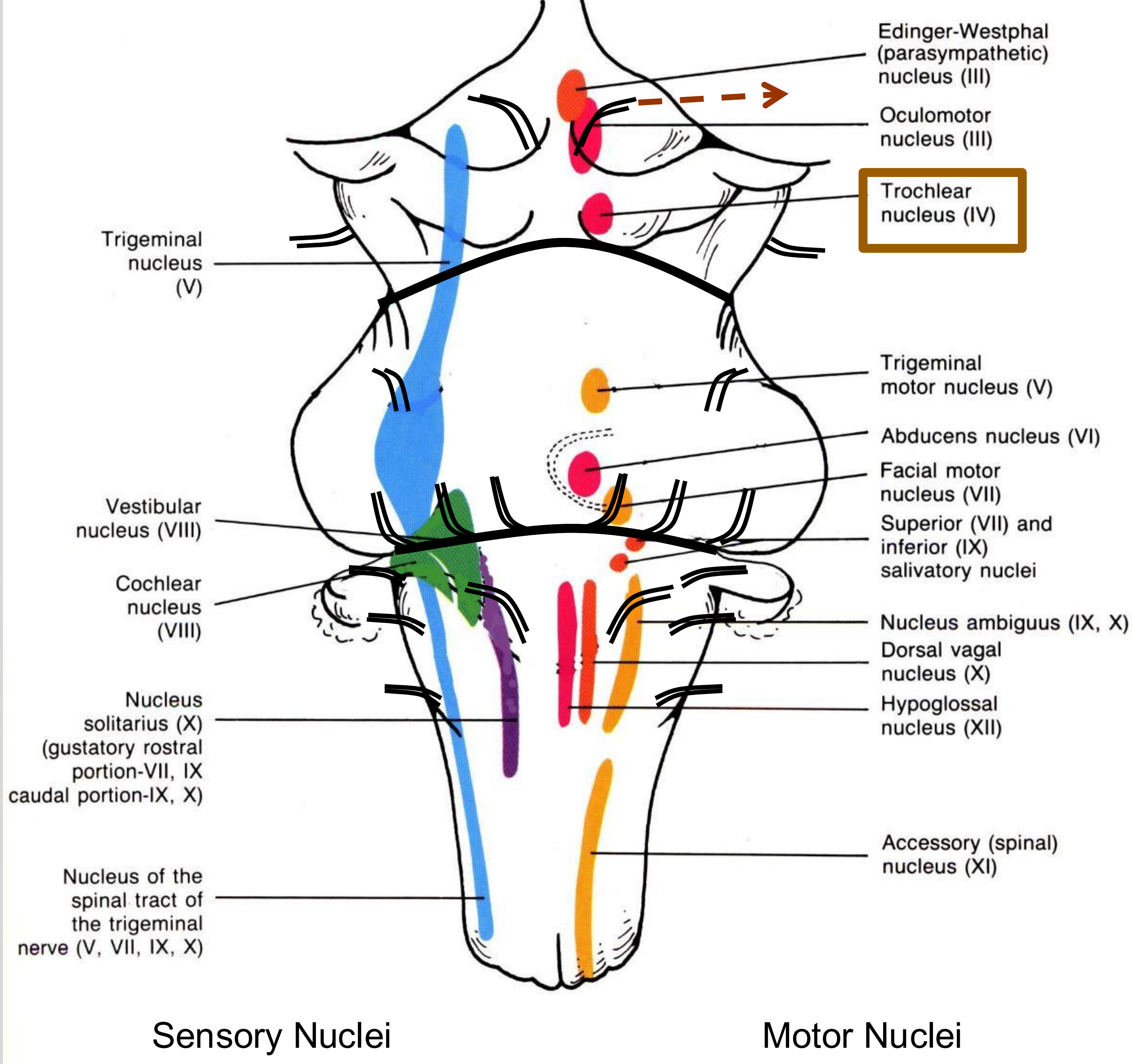
CN IV (trochlear)



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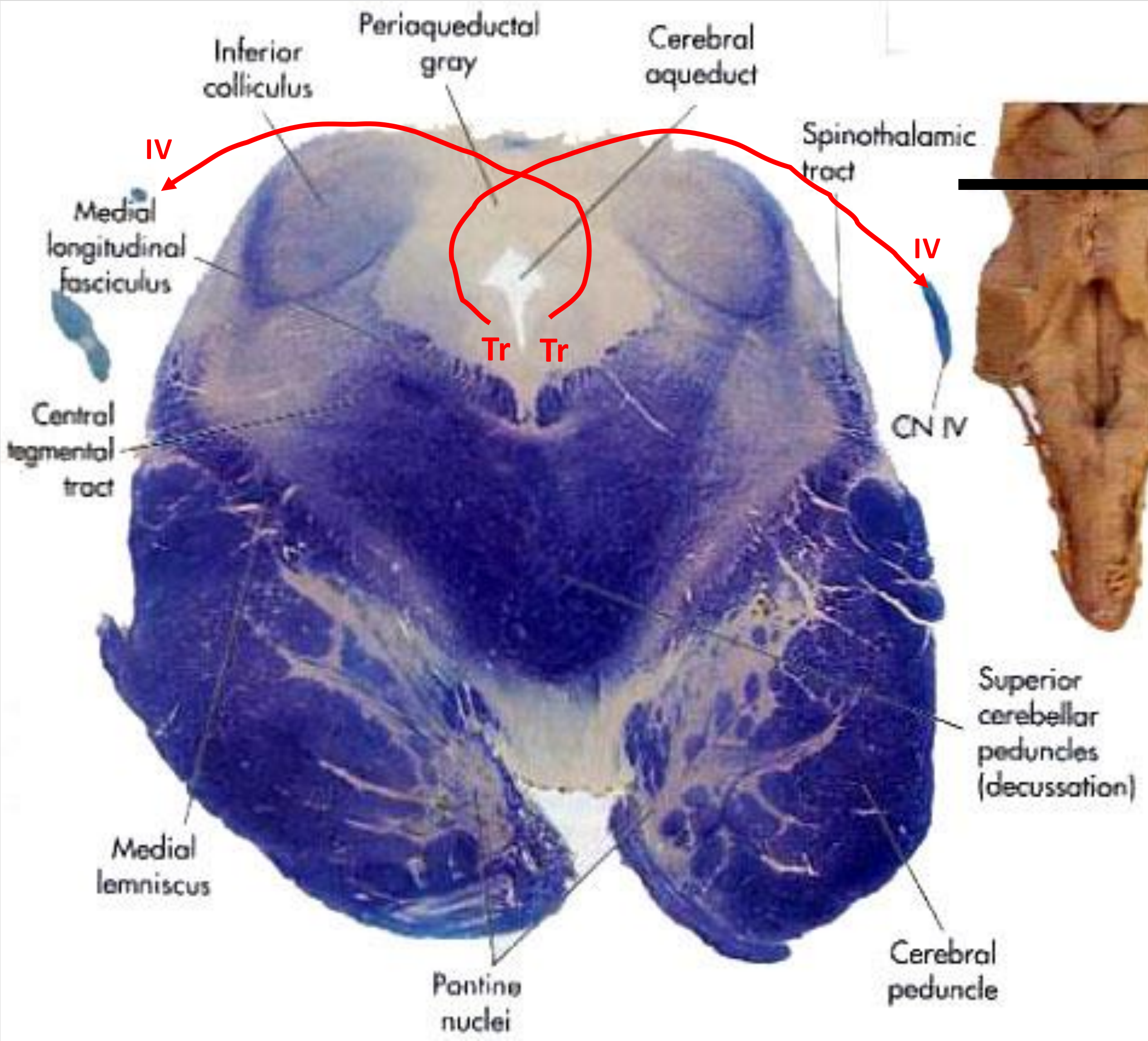
CN IV (trochlear)



caudal midbrain
(inferior colliculus)

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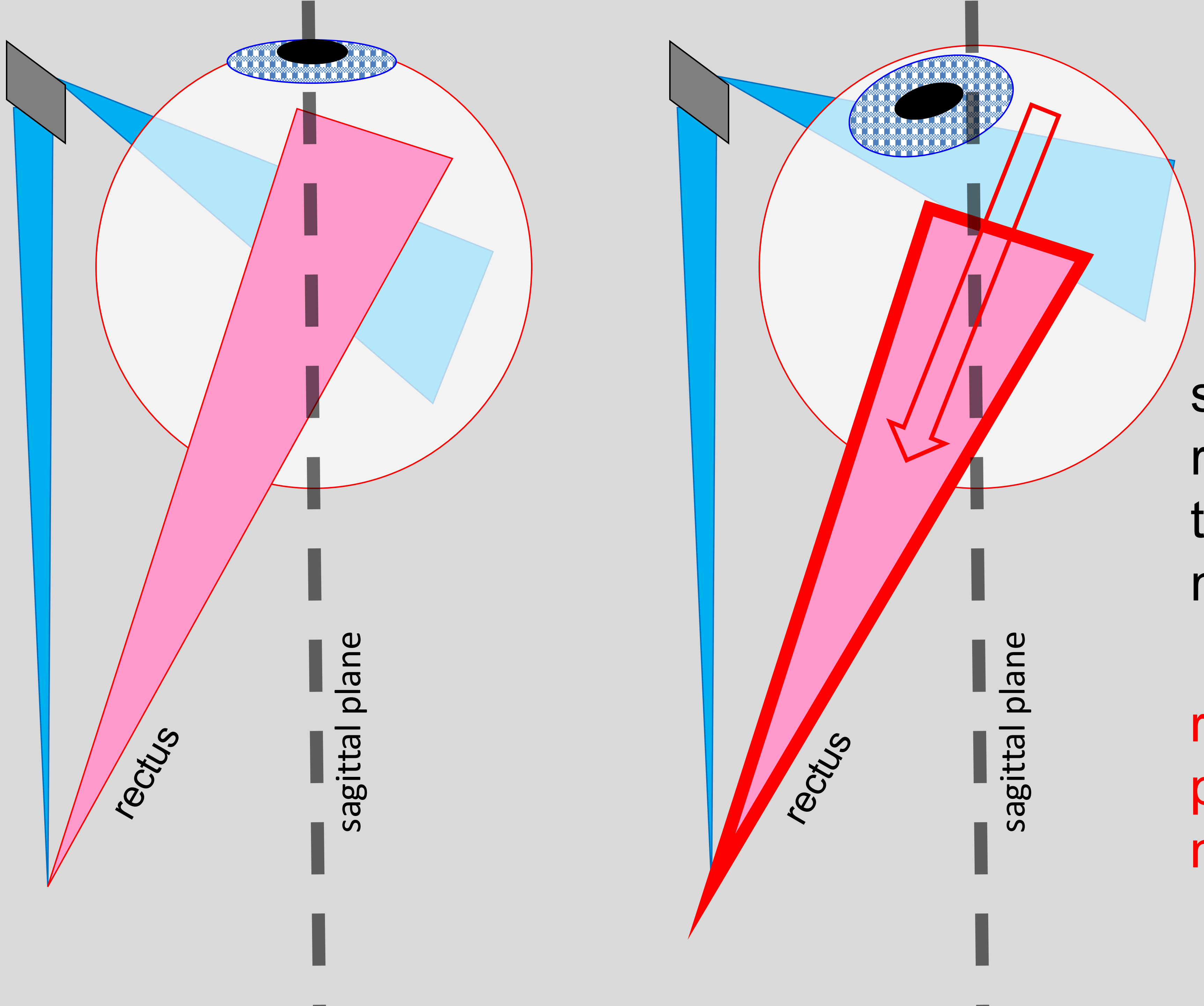
CN IV (trochlear)

IV	**trochlear**	eyes muscles



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superior and inferior rectus muscles attach to orbit ***out*** of eye's midsagittal plane

rectus muscles rotate pupil ***out*** of eye's midsagittal plane

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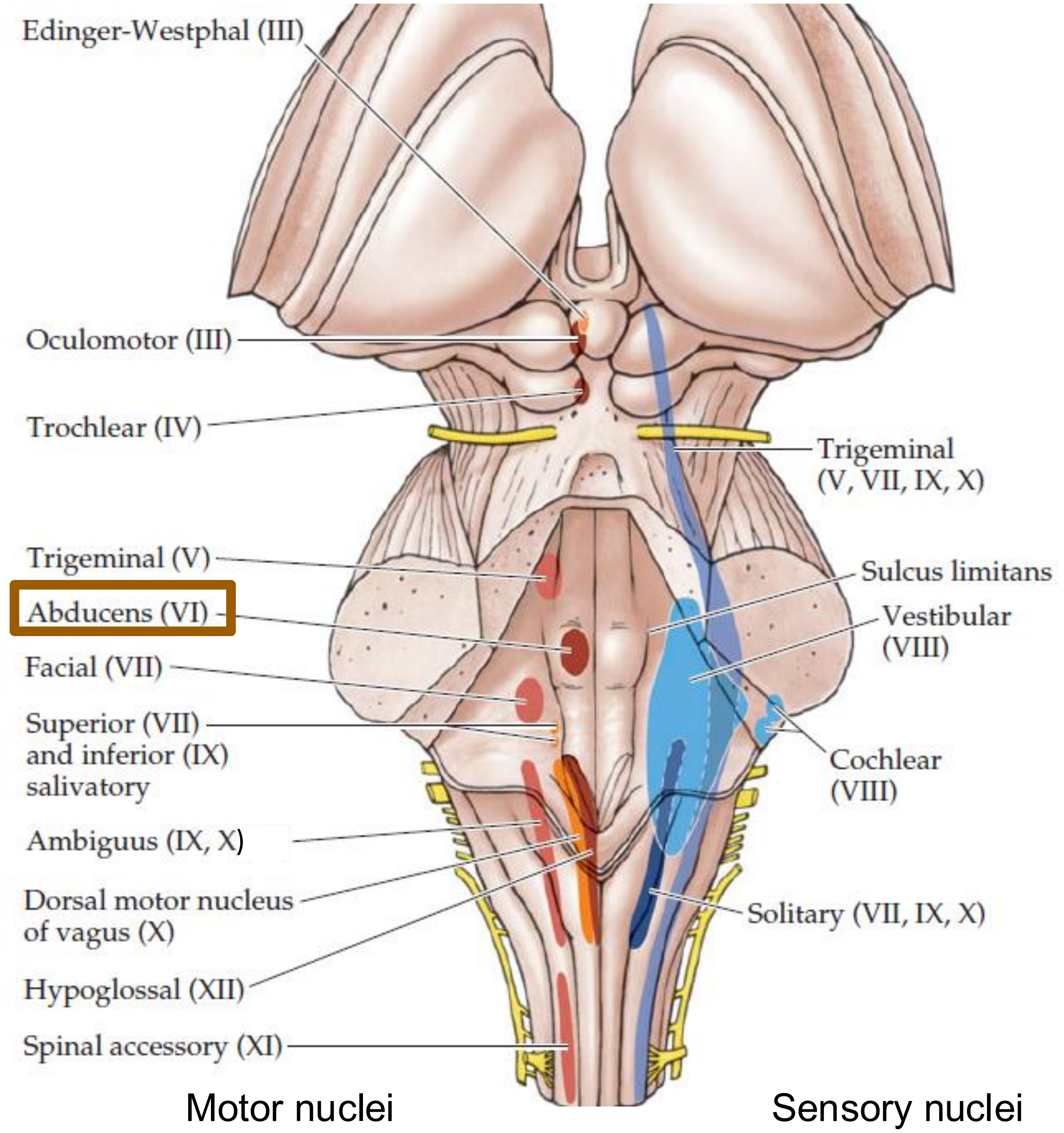
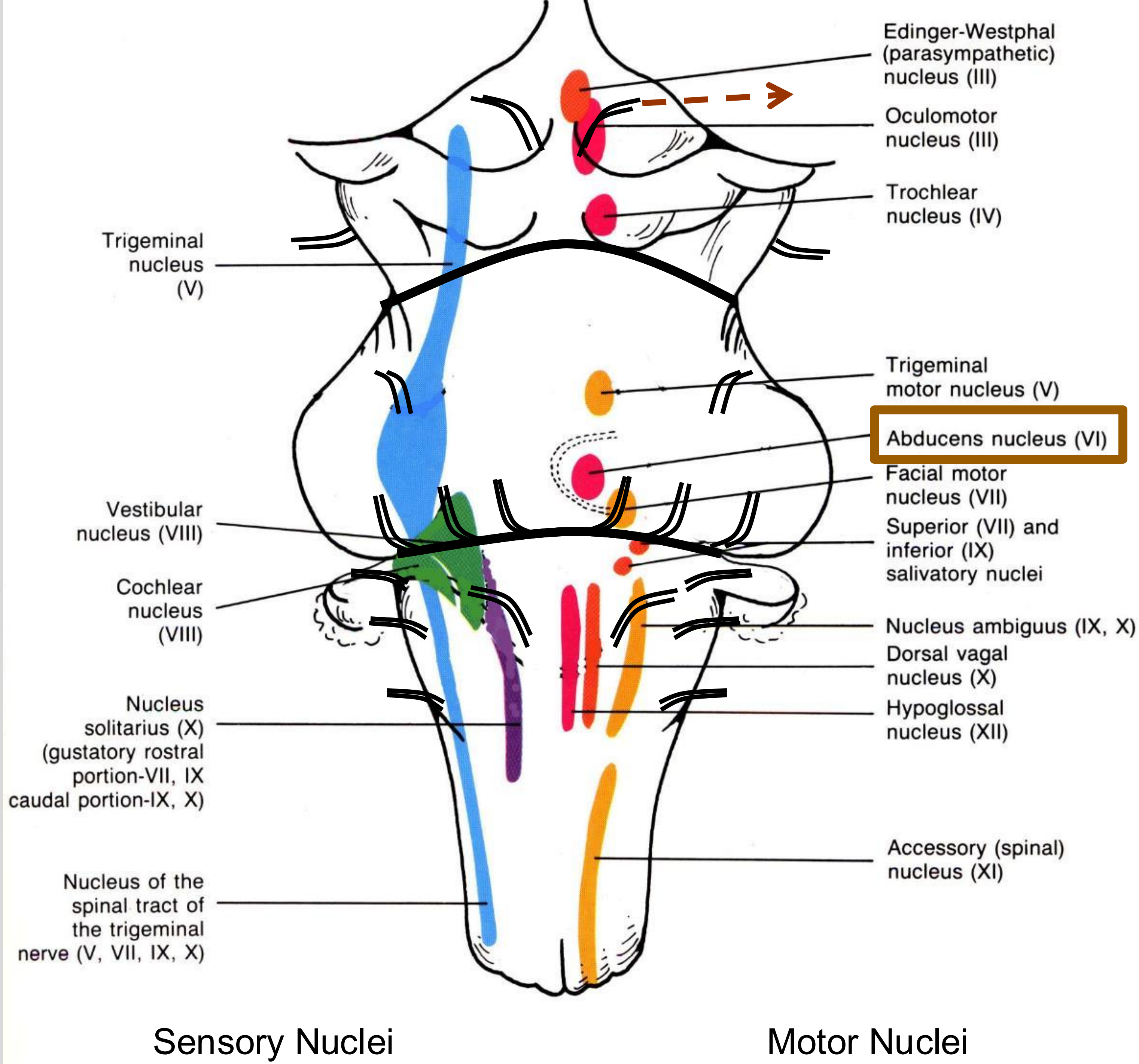
CN VI (abducens)





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CN VI (abducens)

CN VI (abducens)

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SOMATIC MOTOR : ocular positioning

abducens nuclei →

eyes:

lateral rectus: abduction (away from midline)

CN VI paralysis is the most common isolated muscle palsy

Wide range of causes including trauma and diabetic neuropathy

Affected eye pulls medially "esotropia", eso → inner

Horizontal Diplopia (side-by-side double vision)

Treatment

Glasses correcting refractive error (light path)

Eye muscle surgery

CN VI (abducens)

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SOMATIC MOTOR : ocular positioning

abducens nuclei →

eyes:

lateral rectus: abduction (away from midline)

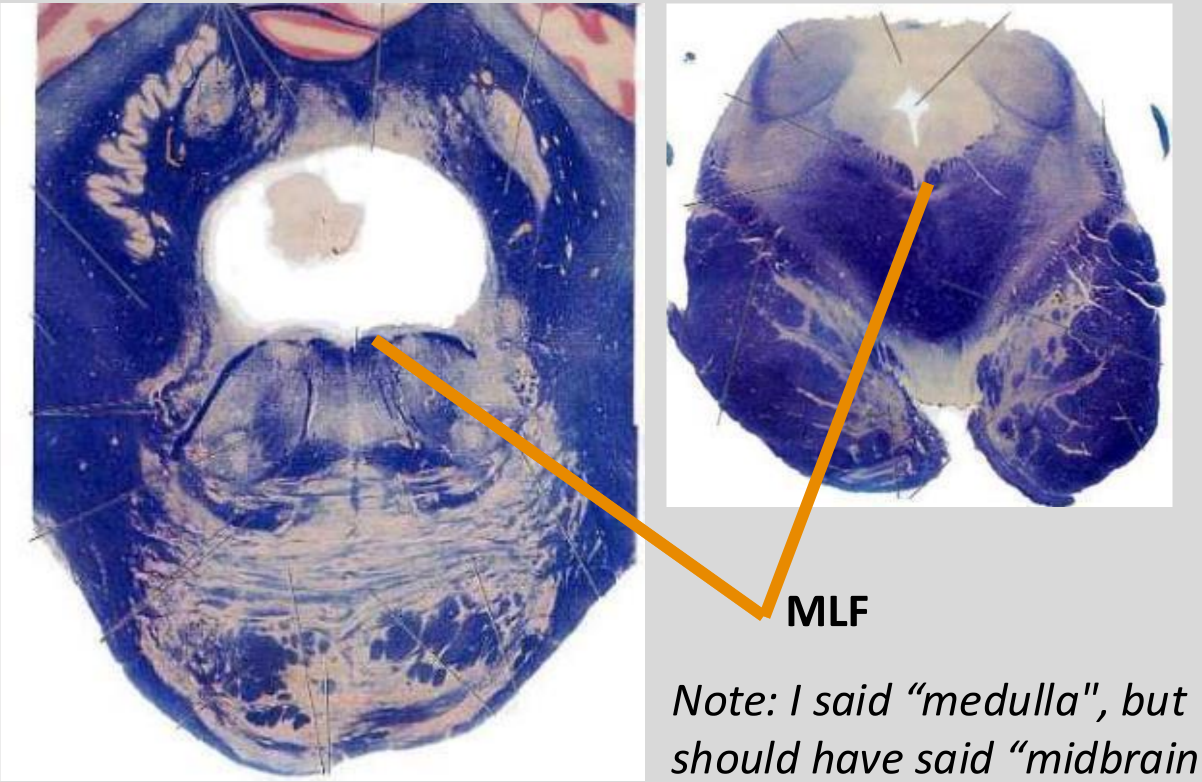
ALSO → contralateral oculomotor nucleus via **M**edial **L**ongitudinal **F**asciculus (not CN VI) to coordinate conjugate lateral gaze (both eyes in same external direction)

MLF bilateral tracts travel between upper medulla to midbrain. Decussations occur immediately before entering.

Impacted by multiple sclerosis, brainstem strokes, head trauma and Arnold-Chiari malformations.

Pons

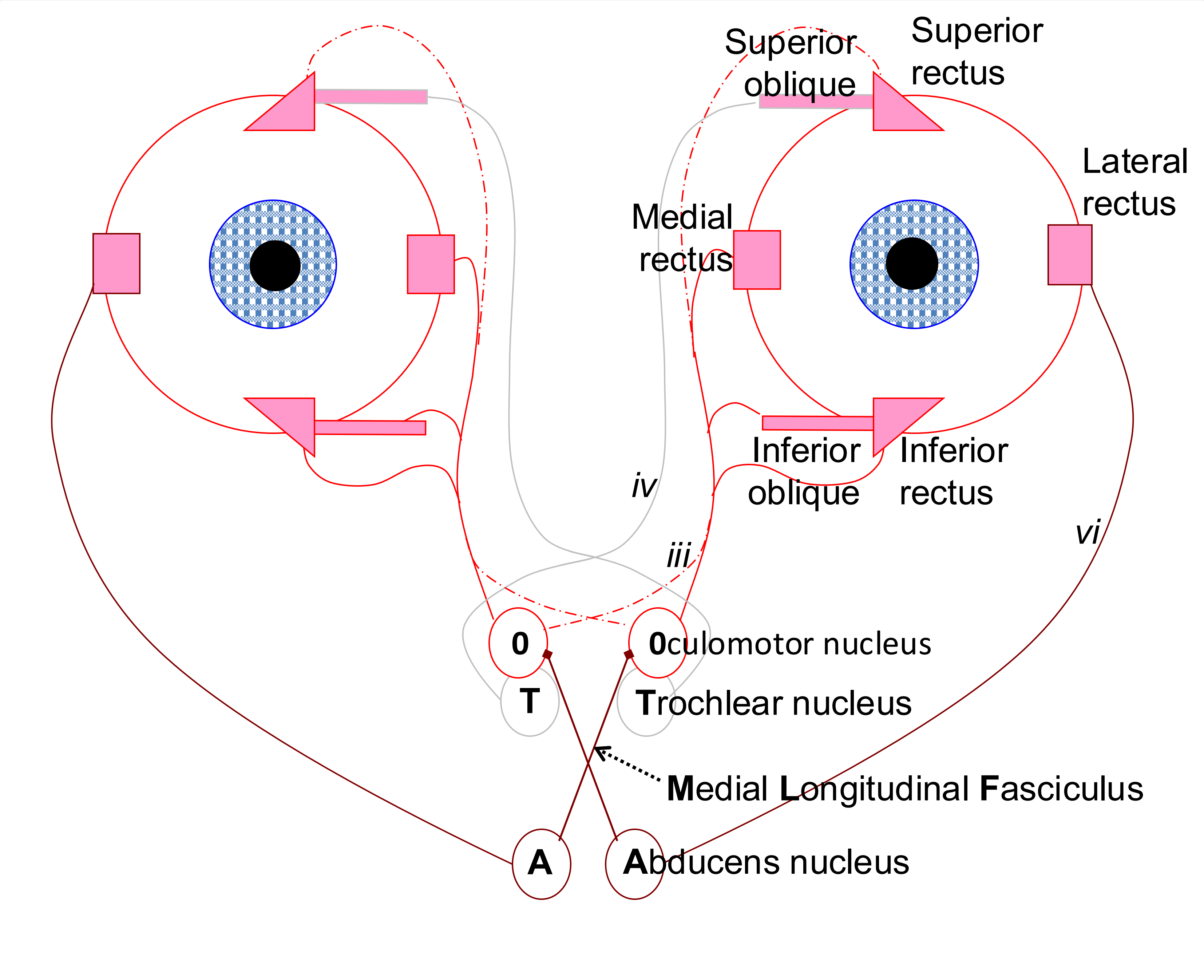
Midbrain



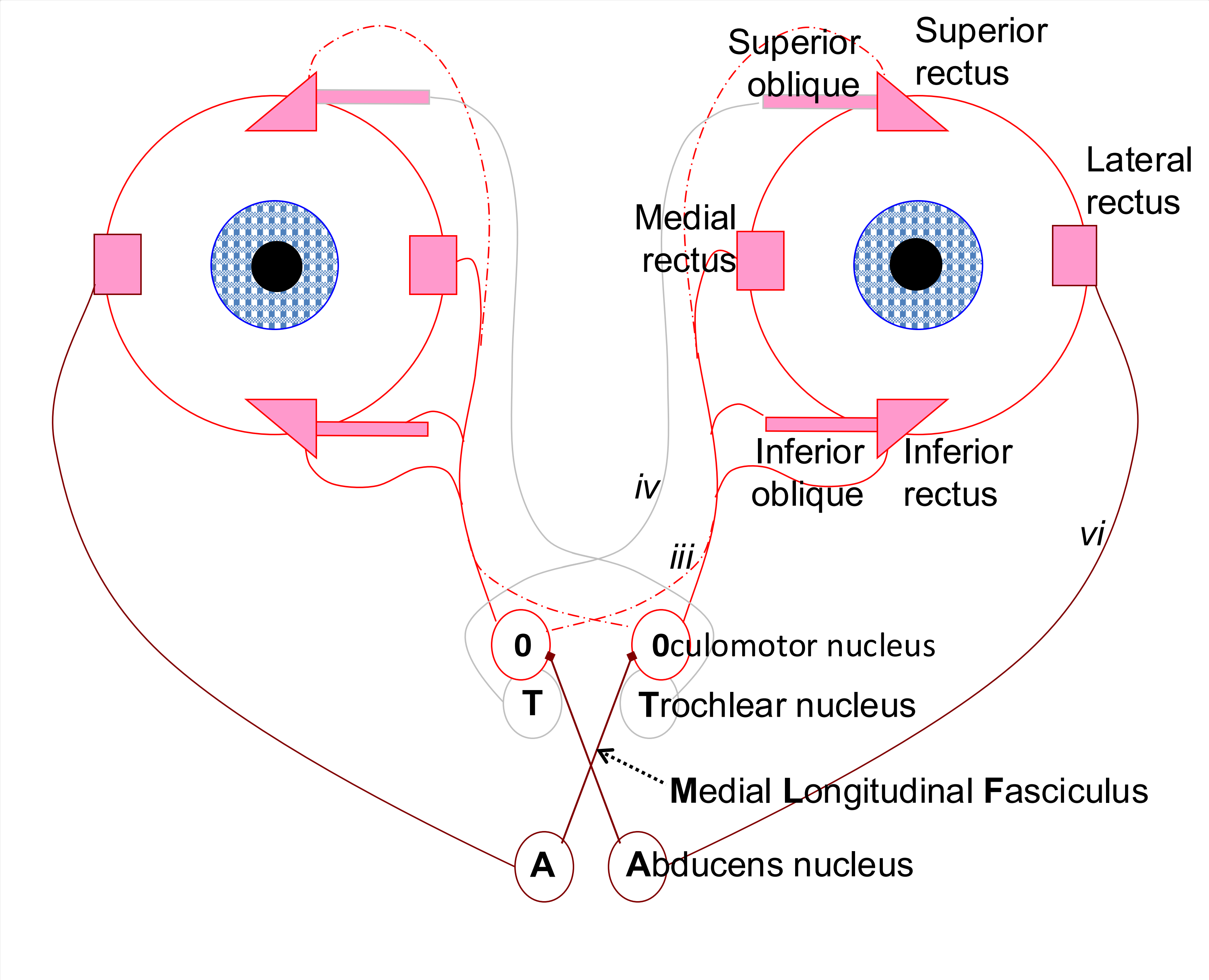
MLF

Note: I said “medulla", but should have said “midbrain“.

Abducens nucleus drives oculomotor nucleus via MLF



CN III, IV, VI control eye movements



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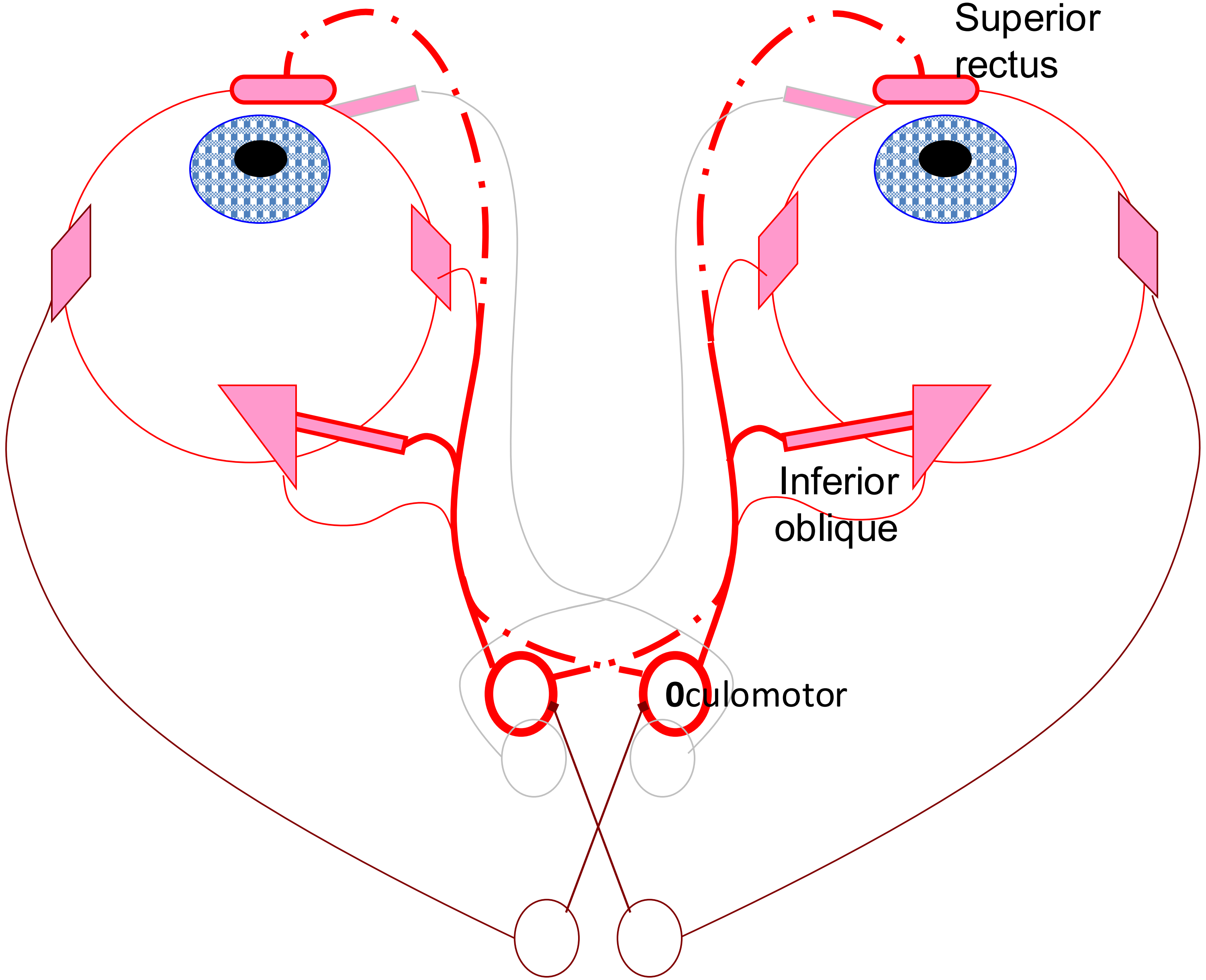
Elevation by CN III



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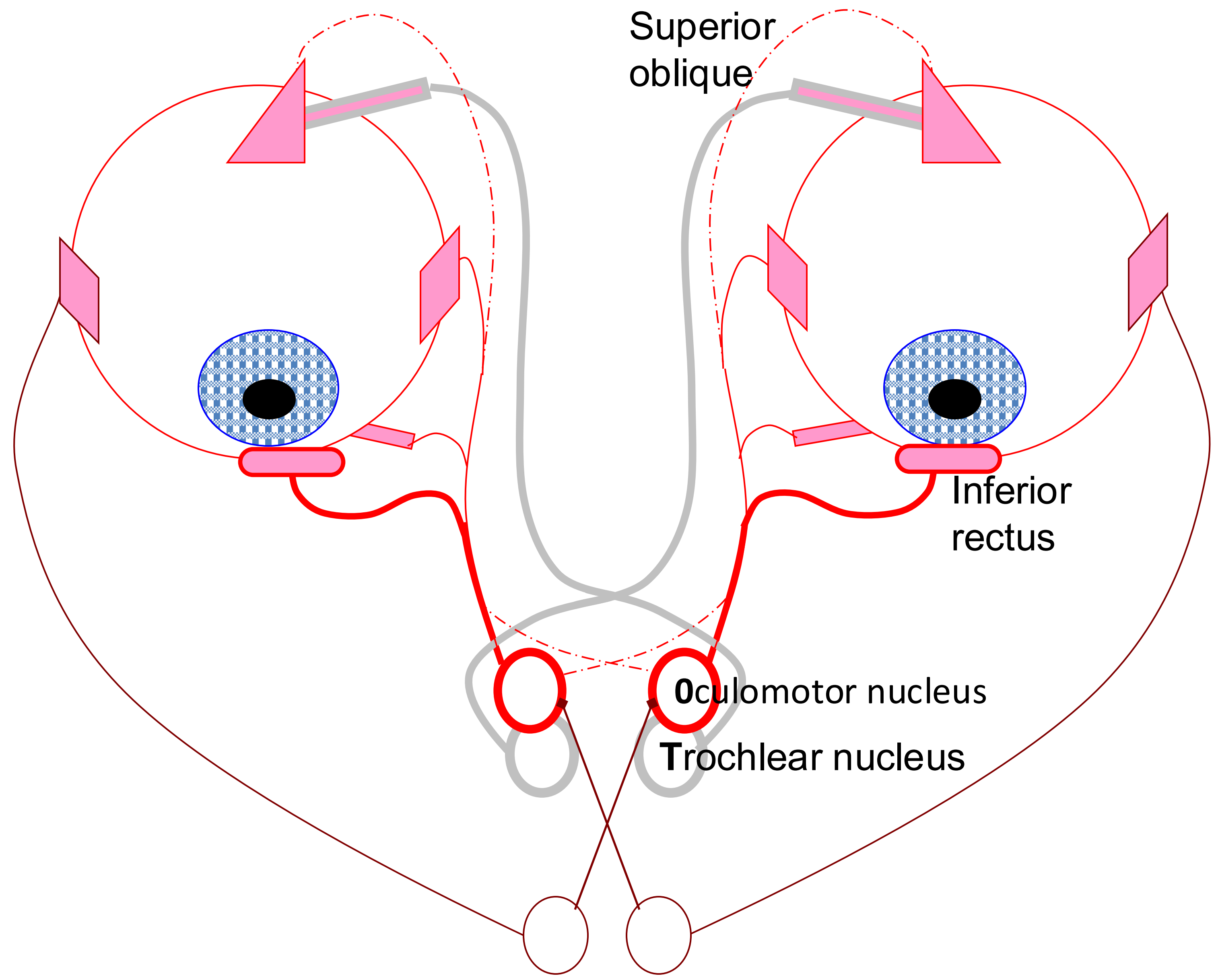
Depression by CN III & IV



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Summary

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- Smell is conveyed by the olfactory tract.
- Vision is conveyed by optic tract, projects thalamus, superior colliculus, and pretectal area.
- Pretectal area bilaterally innervates the Edinger-Westphal nuclei and enables “consensual” light reflex.
- The mechanical action of the different eye muscles requires coordination among the vertical recti and opposing obliques.
- The oculomotor nucleus controls most eye muscles (4 of 6): the contralateral superior rectus, and the ipsilateral medial rectus, inferior rectus, and inferior oblique.
- The superior branch of the oculomotor nerve is from contralateral nucleus for the superior rectus and eyelid. Its inferior branch is from the ipsilateral nucleus for the other eye muscles, pupil and lens.
- The abducens nucleus controls the ipsilateral lateral rectus and contralateral oculomotor nucleus.
- Conjugate lateral gaze requires the medial longitudinal fasciculus.
- Trochlear nucleus controls the contralateral superior oblique.